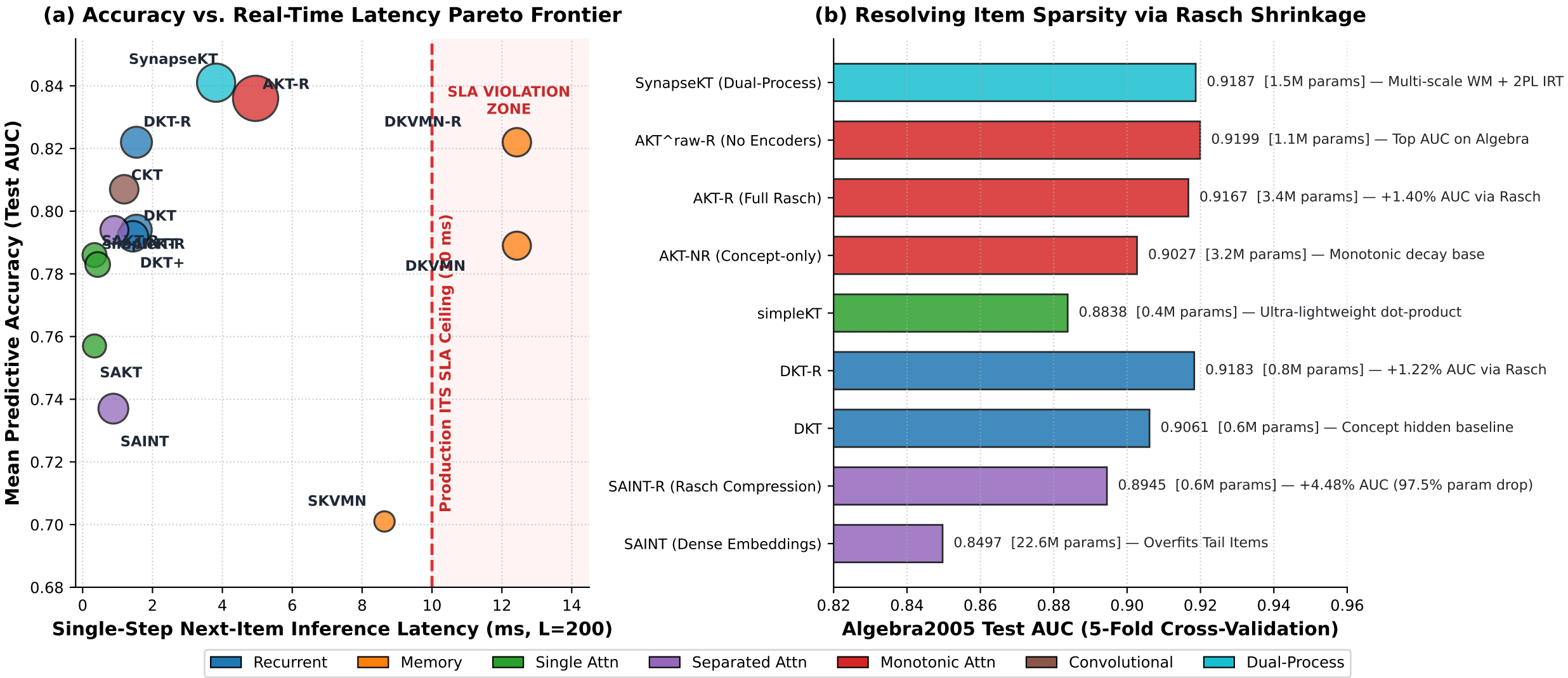
(a) Accuracy vs. Real-Time Latency Pareto Frontier
Mean Predictive Accuracy (Test AUC)
Single-Step Next-Item Inference Latency (ms, L=200)
SynapseKT
AKT-R
DKT-R
CKT
DKT
DKT+
SAKT
SAINT
SKVMN
DKVMN-R
DKVMN
SLA VIOLATION ZONE
Production ITS SLA Ceiling (10 ms)
0.84
0.82
0.80
0.78
0.76
0.74
0.72
0.70
0.68
0
2
4
6
8
10
12
14
(b) Resolving Item Sparsity via Rasch Shrinkage
SynapseKT (Dual-Process)
0.9187 [1.5M params] — Multi-scale WM + 2PL IRT
AKT^raw-R (No Encoders)
0.9199 [1.1M params] — Top AUC on Algebra
AKT-R (Full Rasch)
0.9167 [3.4M params] — +1.40% AUC via Rasch
AKT-NR (Concept-only)
0.9027 [3.2M params] — Monotonic decay base
simpleKT
0.8838 [0.4M params] — Ultra-lightweight dot-product
DKT-R
0.9183 [0.8M params] — +1.22% AUC via Rasch
DKT
0.9061 [0.6M params] — Concept hidden baseline
SAINT-R (Rasch Compression)
0.8945 [0.6M params] — +4.48% AUC (97.5% param drop)
SAINT (Dense Embeddings)
0.8497 [22.6M params] — Overfits Tail Items
0.82
0.84
0.86
0.88
0.90
0.92
0.94
0.96
Algebra2005 Test AUC (5-Fold Cross-Validation)
Recurrent
Memory
Single Attn
Separated Attn
Monotonic Attn
Convolutional
Dual-Process